CV MIDTERM STUDIO

Computer Vision Practice Exams

The supplied original sample and four new variants.

Every exam: 120 minutes, 100 points, five main questions.

Part I contains the question papers. Part II contains model answers, worked steps and partial-credit rubrics.

Mocks 2–3 follow the sample's structure. Mocks 4–5 broaden to the lecture material.

Written work uses self-assessment. Technically sound alternatives may earn credit when they meet the stated requirement.

Part I — Question papers

SUPPLIED ORIGINAL

Original sample — Midterm Variant 1

The supplied sample midterm: its question wording, numbers, code, tables, and marks are preserved. Worked answers and marking suggestions were added for preparation; they are not an official answer key.

120 minutes · 100 points

Name: _____ Group: _____

> Instructions: This is an individual, handwritten, closed-book exam. Show calculations where required. Answers without explanation may receive partial credit. Phones, laptops, smart watches, notes, and AI tools are not permitted.

Question 1 - Image Processing & OpenCV [20 points]

A student is building a system to classify healthy and diseased plant leaves. The collected images have different resolutions, lighting conditions, and some contain image noise.

```
img = cv2.imread("leaf.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_RGB2GRAY)
blur = cv2.GaussianBlur(gray, (4, 4), 0)
```

a) [6 pts] Propose three preprocessing operations that could be useful. For each operation, explain why you would use it.

---

b) [4 pts] The student converts every image to grayscale. Give one advantage and one disadvantage of doing this for plant-disease classification.

---

c) [5 pts] Consider the following code. Identify two problems and provide the corrected code.

---

d) [5 pts] A student says: "Applying more preprocessing always improves classification accuracy." Do you agree? Explain using one example.

Question 2 - Image Classification [20 points]

A classifier produces the following scores:

Image	Cat	Dog	Bird	True class
A	2.4	0.7	-0.2	Cat
B	0.3	1.2	1.8	Dog
C	-0.5	0.4	2.1	Bird
D	1.4	1.6	0.2	Dog

a) [4 pts] Write the predicted class for A, B, C, and D.

---

b) [4 pts] Calculate classification accuracy. Show your calculation.

---

c) [4 pts] Which image was incorrectly classified?

---

d) [4 pts] Explain the difference between a class score and a class probability.

---

e) [4 pts] The model achieves 95% training accuracy but 62% test accuracy. What is the likely problem? Give one possible solution.

Question 3 - Linear Classifier [25 points]

Consider a three-class linear image classifier: s = Wx + b
x = [2, -1]T
W = [[1, 2], [-1, 1], [2, -1]]
b = [0, 1, -1]T
Classes: 0 = Cat, 1 = Dog, 2 = Bird

a) [12 pts] Calculate s = Wx + b. Show every step.

---

b) [3 pts] What class does the classifier predict?

---

c) [5 pts] Explain in your own words what W and b represent.

---

d) [5 pts] Why can't a single linear classifier correctly separate every possible dataset? You may include a small drawing.

Question 4 - Understand the CV Pipeline [20 points]

A student wrote the following image-processing pipeline:

```
img = cv2.imread("cat.jpg")
img = cv2.resize(img, (224, 224))
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
blur = cv2.GaussianBlur(gray, (5, 5), 0)
edges = cv2.Canny(blur, 100, 200)
```

a) [5 pts] Write the dimensions of img after resizing.

---

b) [5 pts] What happens to the number of channels after BGR2GRAY?

---

c) [5 pts] Explain why Gaussian blur might be applied before Canny edge detection.

---

d) [5 pts] Suppose the classification task depends strongly on object color. Would you feed edges alone to the classifier? Explain your decision.

Question 5 - Design Your Own CV System [15 points]

You need to develop a system that classifies waste into Plastic | Paper | Glass | Other. You have collected 2,000 images.

a) [3 pts] How would you organize the dataset?

---

b) [3 pts] Propose a train/test split and calculate the number of images in each part.

---

c) [3 pts] Give two preprocessing operations you would consider.

---

d) [3 pts] What metric would you use to evaluate your classifier? Explain why.

---

e) [3 pts] Your model performs very well on your dataset but poorly when tested using photos taken by another student's phone. Give one possible reason.

NEW PRACTICE VARIANT

Mock 2 — Sample-style: traffic signs and fruit sorting

New practice questions closely matching the sample structure and mark distribution. They are original preparation material, not a prediction of the real exam. For arrays, state (height, width, channels); classify by the largest score.

120 minutes · 100 points

Name: _____ Group: _____

Question 1 - Image Processing & OpenCV [20 points]

A traffic-sign classifier receives photographs of different sizes, with shadows and mild sensor noise. Sign color helps distinguish several categories.

```
img = cv2.imread("sign.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_RGB2GRAY)
blur = cv2.GaussianBlur(gray, (6, 6), 0)
```

a) [6 pts] Suggest three preprocessing operations. Explain the purpose of each in this scenario.

b) [4 pts] Give one advantage and one risk of using grayscale traffic-sign images.

c) [5 pts] Identify two problems in the code and write a corrected version.

d) [5 pts] Would very strong smoothing necessarily improve this classifier? Use a specific sign feature to explain.

Question 2 - Image Classification [20 points]

Use the largest score to predict the class for each sign image.

Image	Stop	Yield	Speed limit	True class
A	1.2	0.1	-0.8	Stop
B	-0.4	2.1	0.7	Yield
C	0.8	1.9	1.5	Speed limit
D	-1.1	0.2	1.0	Speed limit

a) [4 pts] Write the predictions for A, B, C, and D.

---

b) [4 pts] Calculate accuracy and show the fraction used.

---

c) [4 pts] Identify every incorrectly classified image and give its predicted and true class.

---

d) [4 pts] Can a raw score of 2.1 be read as a 210% probability? Explain.

---

e) [4 pts] Training accuracy is 98%, while test accuracy is 66%. Name a likely problem and one remedy.

Question 3 - Linear Classifier [25 points]

s = Wx + b; x = [3, 2]T; W = [[1, -1], [-2, 2], [0.5, 1]]; b = [1, -1, 0]T. Classes: 0 = Stop, 1 = Yield, 2 = Speed limit.

a) [12 pts] Calculate all three scores. Show the dot products and bias additions.

---

b) [3 pts] Which class is predicted, and why?

---

c) [5 pts] Describe the shapes of W and b and what each row of W means.

---

d) [5 pts] Can a two-class linear classifier perfectly classify the XOR pattern in the original two input coordinates? Explain.

Question 4 - Understand the CV Pipeline [20 points]

Assume sign.jpg loads successfully as a color image. OpenCV resize uses (width, height).

```
img = cv2.imread("sign.jpg")
img = cv2.resize(img, (160, 96))
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
blur = cv2.GaussianBlur(gray, (3, 3), 0)
edges = cv2.Canny(blur, 50, 150)
```

a) [5 pts] Give the NumPy shape of img after the resize, using (height, width, channels).

---

b) [5 pts] Give the channel counts and spatial array shapes of gray and edges.

---

c) [5 pts] Why place Gaussian blur before Canny rather than rely on an unsmoothed noisy image?

---

d) [5 pts] If red and blue signs have similar silhouettes, is an edge map alone a sufficient representation? Explain one improvement.

Question 5 - Design a Fruit-Sorting System [15 points]

You have 3,200 labeled photos for Apple | Banana | Orange | Other. Some fruits were photographed several times.

a) [3 pts] How would you organize images and labels while preventing an unrealistically easy test?

---

b) [3 pts] Use an 80%/20% train/test split. Calculate the image counts.

---

c) [3 pts] Give two preprocessing choices suited to fruit classification.

---

d) [3 pts] If Other has far fewer examples, what evaluation metric would you report and why?

---

e) [3 pts] Performance falls on images taken in a supermarket rather than your studio. Give one plausible reason.

NEW PRACTICE VARIANT

Mock 3 — Sample-style: industrial inspection

A second new practice variant matching the sample's five sections and marks. It changes the numbers and code issues. Assume image reads succeed unless a question asks about file handling.

120 minutes · 100 points

Name: _____ Group: _____

Question 1 - Image Processing & OpenCV [20 points]

A factory classifies metal bolts as Normal | Crack | Rust. Images have different sizes, lighting varies, and some contain salt-and-pepper noise.

```
img = cv2.imread("bolt.png", cv2.IMREAD_GRAYSCALE)
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
blur = cv2.medianBlur(gray, 4)
```

a) [6 pts] Suggest three preprocessing operations and explain each choice.

---

b) [4 pts] Give one benefit and one limitation of grayscale images for this task.

---

c) [5 pts] Find the two issues in the code and provide a corrected version.

---

d) [5 pts] Explain why removing all small dark regions could damage crack detection.

Question 2 - Image Classification [20 points]

The model outputs raw scores. Use argmax to classify each image.

Image	Normal	Crack	Rust	True class
A	1.4	0.2	-0.1	Normal
B	0.7	0.5	1.9	Rust
C	1.8	1.5	0.2	Crack
D	-0.3	0.9	1.1	Crack

a) [4 pts] Give the predictions for A, B, C, and D.

---

b) [4 pts] Calculate accuracy with your working.

---

c) [4 pts] Which images are incorrect? Give their true and predicted labels.

---

d) [4 pts] Why is a negative class score allowed, while a negative class probability is not?

---

e) [4 pts] Training accuracy is 96% and test accuracy is 63%. Give one plausible diagnosis and a matched improvement.

Question 3 - Linear Classifier [25 points]

s = Wx + b; x = [-2, 3]T; W = [[2, 1], [-1, 2], [1, -2]]; b = [0, -2, 1]T. Classes: 0 = Normal, 1 = Crack, 2 = Rust.

a) [12 pts] Calculate all scores, showing the row-by-row products and biases.

---

b) [3 pts] State the predicted class and decision rule.

---

c) [5 pts] Explain how changing one row of W differs from changing one entry of b.

---

d) [5 pts] Why might a linear model fail if Rust examples form a ring around Normal examples in two feature dimensions?

Question 4 - Understand the CV Pipeline [20 points]

Assume bolt.jpg is successfully read as a uint8 color image. THRESH_BINARY produces maxval when the input exceeds the threshold and 0 otherwise.

```
img = cv2.imread("bolt.jpg")
img = cv2.resize(img, (128, 64))
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
_, binary = cv2.threshold(gray, 120, 255, cv2.THRESH_BINARY)
```

a) [5 pts] Give img.shape after resize, in (height, width, channels) order.

---

b) [5 pts] Give the shapes and possible pixel values of gray and binary.

---

c) [5 pts] Why can the fixed threshold 120 fail when the room lighting changes?

d) [5 pts] Would binary shape information alone be enough to distinguish orange rust from gray metal? Explain one alternative.

Question 5 - Design an Inspection System [15 points]

You have 1,500 metal-part images for Normal | Cracked | Rusted | Other. Photos from the same physical part may be very similar.

a) [3 pts] Describe a suitable label organization and one rule for independent evaluation.

b) [3 pts] Use 70% train, 15% validation, and 15% test. Calculate counts.

c) [3 pts] Name two useful preprocessing operations for this inspection task.

d) [3 pts] If missing a crack is expensive, what per-class metric matters and why?

e) [3 pts] The system fails on a new conveyor belt although the parts are similar. Give one plausible cause.

NEW PRACTICE VARIANT

Mock 4 — Expanded lecture coverage: losses, learning, CNNs

Expanded preparation beyond the supplied sample: Lecture 2 kNN, Lecture 3 losses, Lecture 4 derivatives and updates, and Lecture 5 CNN shapes. All logarithms are natural; include one bias per convolutional output channel. These sections broaden coverage and are not a forecast of the exam.

120 minutes · 100 points

Name: _____ Group: _____

Question 1 - kNN and Evaluation [20 points]

A query has features x = [1,2]. Use L1 distance d(x,z)=|x1-z1|+|x2-z2|. The training table follows. In a distance tie, use alphabetical ID order; for a vote tie, choose the label of the closest selected neighbor.

ID	Features	Label
A	[0,0]	Plastic
B	[2,2]	Paper
C	[1,4]	Paper
D	[4,1]	Glass
E	[0,2]	Paper

a) [6 pts] Calculate the L1 distance to every training point.

b) [4 pts] Predict using k=1 and k=3. Identify the selected neighbors.

c) [5 pts] Where should you choose k and the distance metric? Why should the final test not be used for that choice?

d) [5 pts] Explain one difficulty of raw-pixel nearest neighbors on high-resolution images and one possible improvement.

Question 2 - Scores and Losses [20 points]

For one example, scores in class-index order are s=[2,1,0] and the true class index is y=1. Softmax pj=exp(sj)/sum_k exp(sk); cross-entropy LCE=-ln(py). Multiclass hinge loss is LH=sum_{j≠y} max(0,sj-sy+1). No regularization is included.

a) [4 pts] Give the predicted class and say whether it is correct.

---

b) [6 pts] Calculate all softmax probabilities and the cross-entropy loss. Round to three decimals.

---

c) [6 pts] Calculate the multiclass hinge loss, showing each incorrect-class term.

---

d) [4 pts] If the true class has the highest score, must either loss equal zero? Explain the two conditions.

Question 3 - Gradient Descent and Branching [25 points]

One training example has x=2, target t=1, prediction yhat=wx+b, and L=0.5(yhat-t)^2. Initially w=0.25, b=0. Use learning rate η=0.1 and simultaneous gradient-descent updates of w and b using the old values. There is no regularization.

a) [5 pts] Calculate the initial prediction and loss.

---

b) [8 pts] Find $\partial L/\partial w$ and $\partial L/\partial b$, showing the chain rule.

---

c) [6 pts] Perform one update, then calculate the new prediction and loss on the same example.

---

d) [6 pts] For the separate graph $f(u)=u\cdot u+3\cdot u$ at $u=2$, find df/du. Explain why gradients from both branches must be added.

Question 4 - CNN Shapes and Parameters [20 points]

Input shape is 32×32×3 (H×W×C). Layer 1: convolution with eight 3×3 filters, stride 1, padding 1, one bias/filter, followed by ReLU. Then max pool 2×2, stride 2, no padding. Layer 2: sixteen 3×3 filters, stride 1, no padding, one bias/filter. Use $O=\text{floor}((N-K+2P)/S)+1$ for each spatial axis.

a) [6 pts] Find the shape after layer 1/ReLU and after max pooling.

---

b) [5 pts] Find the output shape of layer 2.

---

c) [5 pts] Count trainable parameters in each convolutional layer, including biases.

---

d) [4 pts] Explain how local connectivity and weight sharing help compared with a fully connected layer on flattened pixels.

Question 5 - Build and Evaluate a Leaf Classifier [15 points]

A leaf dataset contains 7,500 images, including several views from each plant. A mobile app will classify Healthy | Disease A | Disease B.

a) [5 pts] Use a nominal 70%/15%/15% train/validation/test allocation. Calculate image counts and explain how repeated views should be handled.

---

b) [5 pts] The CNN reaches 99% training accuracy but 68% validation accuracy. Suggest a cause and two actions, with reasons.

---

c) [5 pts] Name one suitable architecture family for a phone and explain how you would compare it with a larger model.

NEW PRACTICE VARIANT

Mock 5 — Expanded lecture coverage: filtering, backprop, architecture

Expanded practice using Szeliski Chapter 3 image processing and Lectures 3–5. Numerical conventions are explicit. CNN calculations use cross-correlation as in deep-learning libraries, one bias per output channel, and exact integer shapes.

120 minutes · 100 points

Name: _____ Group: _____

Question 1 - Image Processing Decisions [20 points]

Work with scalar grayscale intensities. For the first two subparts, calculate only the center output from the stated 3×3 patch; no border rule is needed.

a) [5 pts] For patch [[1,2,3],[4,5,6],[7,8,9]], apply a 3×3 averaging filter with every coefficient 1/9. Calculate the center output.

---

b) [5 pts] For patch [[10,10,11],[9,255,10],[11,9,10]], calculate the 3×3 median-filter center output. Why is this useful for the bright outlier?

---

c) [5 pts] Transform intensities [0,64,128,255] using J=clip(1.5I+20,0,255). Give outputs and explain why clipping should occur before casting to uint8.

---

d) [5 pts] Why can naive downsampling create aliasing, and what processing step can reduce it?

Question 2 - A Two-Layer Network [20 points]

x=[1,-2]T. W1=[[1,1],[-1,-2]], b1=[0.5,-0.5]T; z=W1x+b1; h=ReLU(z). W2=[[2,-1],[1,2]], b2=[0.5,-0.5]T; s=W2h+b2. Classes are 0 and 1, predicted by argmax.

a) [6 pts] Calculate z and h, showing the effect of ReLU.

b) [4 pts] Calculate s and the predicted class.

c) [6 pts] If ReLU were removed, derive the effective single-layer Wprime and bprime. Explain the implication.

d) [4 pts] The first hidden preactivation is negative for this example. What happens to its gradient through ReLU? Does that alone prove the unit is permanently dead?

Question 3 - Matrix Backpropagation [25 points]

s=Wx+b, x=[2,-1]T, W=[[1,0],[-1,2]], b=[0.5,-0.5]T. For this example the upstream gradient is g=$\partial L/\partial s$=[3,-2]T. For subpart (d), use the scalar loss L=3s0-2s1 and vary only w=W[0,0].

a) [5 pts] Calculate s and give the shapes of W, x, and g.

b) [8 pts] Calculate $\partial L/\partial W$ and $\partial L/\partial b$. Show the outer product.

c) [6 pts] Calculate $\partial L/\partial x$ and explain the transpose.

d) [6 pts] At w=1, use h=0.00001, L(w+h)=16.500060, and L(w-h)=16.499940. Compute the central-difference gradient and compare with the analytic result for w.

Question 4 - CNN Architecture [20 points]

Input: 28×28×1. Conv A: six 5×5 filters, S=1, P=0. Max pool: K=2, S=2, P=0. Conv B: twelve 3×3 filters, S=1, P=1. Then global average pooling and a fully connected 3-class head. Every convolution filter and every FC output has one bias. Shapes are H×W×C.

a) [6 pts] Give shapes after Conv A, max pool, Conv B, and global average pooling.

---

b) [6 pts] Count parameters in Conv A, Conv B, and the FC head. Give the total.

---

c) [4 pts] With stride 1, compare the receptive field of two 3×3 convolutions with one 5×5 convolution. Ignore biases and assume every layer has C input/output channels. Compare weight counts.

---

d) [4 pts] State the residual-block relation and explain when its skip connection needs a projection.

Question 5 - Evaluation Under Imbalance and Shift [15 points]

An inspection test set contains 1000 images: 20 faulty and 980 normal. Treat Faulty as positive. Counts: true positives TP=12, false negatives FN=8, false positives FP=28, true negatives TN=952. All model choices were fixed before this final test.

a) [5 pts] Calculate Faulty precision and recall.

---

b) [5 pts] Calculate accuracy and compare with always predicting Normal. Why can accuracy alone mislead here?

---

c) [5 pts] Recall worsens on a second phone. Give one possible cause and one remedy or evaluation check. Should you keep adjusting hyperparameters against this final test set?

Part II — Answers and marking guides

Give credit for the reasoning and calculations specified. Open-ended model answers are examples, and do not exclude valid alternatives.

Original sample — Midterm Variant 1 — Key

Question 1 - Image Processing & OpenCV [20 points]

a) [6 pts]

Answer: Resize to a common input size, preferably preserving the aspect ratio with padding, so batches have consistent dimensions. Apply a suitable illumination or intensity adjustment when lighting differs, while preserving disease-related color. Use mild Gaussian or median denoising when noise is present, choosing the filter for the noise type. Evaluate the choices on validation data.

1. Identify the stated problems: resolution, lighting, and noise.
2. Map each proposed operation to one problem.
3. Check that preprocessing preserves lesion texture and useful color.

Marking guide

- 2 points per valid operation with a relevant reason: 1 for the operation and 1 for the explanation.
- Accept alternatives such as crop/segmentation, normalization, or a justified contrast adjustment; no single trio is required.

b) [4 pts]

Answer: Advantage: one channel reduces storage and computation and can reduce irrelevant color variation. Disadvantage: yellowing, discoloration, or colored lesions may distinguish disease classes; grayscale removes that color information.

1. Compare three color channels with one intensity channel.
2. Connect lost color information to this specific task.

Marking guide

- 2 points for a valid advantage with explanation.
- 2 points for a task-related disadvantage. Accept any equivalent reason; grayscale is not automatically beneficial.

c) [5 pts]

Answer: cv2.imread loads a color image in BGR order by default, so use COLOR_BGR2GRAY. A specified Gaussian kernel must have positive odd dimensions, so replace (4, 4) with (5, 5), for example. Corrected code:

```
img = cv2.imread("leaf.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
blur = cv2.GaussianBlur(gray, (5, 5), 0)
```

An optional check `if img is None: raise FileNotFoundError("leaf.jpg")` can be placed after imread.

1. Match the conversion constant to the actual BGR input order.
2. Use an odd kernel such as (3, 3) or (5, 5).
3. A file-load check is useful but is additional to the two intended corrections.

Marking guide

- 2 points for explaining BGR versus RGB and correcting the conversion.
- 2 points for identifying the invalid even kernel and replacing it with a positive odd kernel.
- 1 point for coherent corrected code. Accept another valid odd kernel; optional missing-file handling is not required.

d) [5 pts]

Answer: No. Excessive smoothing can erase small lesions or vein texture needed to identify disease. A heavy blur may therefore reduce accuracy even though it removes noise. Preprocessing choices should be tested on validation data using the same evaluation protocol.

1. Reject the word "always".
2. Give an operation that can remove task-relevant information.
3. Explain the resulting classification error.

Marking guide

- 1 point for disagreeing with the universal claim.
- 3 points for a concrete operation and correct explanation of the harm.
- 1 point for connecting the decision to task evidence or validation. Accept color removal, distorted resizing, or other well-explained examples.

Question 2 - Image Classification [20 points]

a) [4 pts]

Answer: A: Cat; B: Bird; C: Bird; D: Dog. Choose the class with the largest score in each row.

1. A: max(2.4, 0.7, -0.2) = 2.4, Cat.
2. B: max(0.3, 1.2, 1.8) = 1.8, Bird.
3. C: max(-0.5, 0.4, 2.1) = 2.1, Bird.
4. D: max(1.4, 1.6, 0.2) = 1.6, Dog.

Marking guide

- 1 point per correct predicted class.

b) [4 pts]

Answer: Accuracy = number correct / number evaluated = 3/4 = 0.75 = 75%. A, C, and D are correct.

1. Compare each predicted class with its true class.
2. Count 3 correct predictions out of 4.
3. Compute 3 ÷ 4 × 100 = 75%.

Marking guide

- 2 points for the correct count and denominator; 1 point for the formula; 1 point for 75% (or 0.75). Exact fraction 3/4 is accepted.

c) [4 pts]

Answer: Image B: its true class is Dog, but Bird has the highest score.

1. Compare B’s largest score, 1.8 for Bird, with its Dog label.

Marking guide

- 2 points for B and 2 for correctly identifying Bird versus Dog.

d) [4 pts]

Answer: A class score is a raw model output used to compare classes; it can be negative or greater than 1 and need not sum to 1. A probability distribution has values in [0,1] that sum to 1. Softmax transforms scores into a probability distribution, but those values are not guaranteed to match real frequencies without calibration.

1. State that raw scores have no probability-range or normalization requirement.
2. State the range and sum requirements for mutually exclusive class probabilities.
3. A score can choose the predicted class without being a probability.

Marking guide

- 2 points for the raw-score explanation; 2 for the probability explanation.
- Accept a correct softmax example. Calibration detail is useful but not required for full marks.

e) [4 pts]

Answer: The gap suggests overfitting: the model fits training-specific patterns but generalizes poorly. One remedy is stronger regularization or appropriate training augmentation, selected using a validation set. Other possibilities, such as different train/test distributions, should be investigated if evidence supports them.

1. Compare training accuracy with accuracy on unseen examples.
2. Name a plausible cause of the generalization gap.
3. Connect a remedy to the cause.

Marking guide

- 2 points for overfitting or another well-justified cause, and 2 for a suitable remedy.
- Accept more representative training data, early stopping on validation, weight decay, valid augmentation, or an appropriately simpler model. The gap alone does not prove overfitting.

Question 3 - Linear Classifier [25 points]

a) [12 pts]

Answer: s = [0, -2, 4]T. Cat: 1·2 + 2·(-1) + 0 = 0. Dog: (-1)·2 + 1·(-1) + 1 = -2. Bird: 2·2 + (-1)·(-1) - 1 = 4.

1. Wx = [2 - 2, -2 - 1, 4 + 1]T = [0, -3, 5]T.
2. Add b = [0, 1, -1]T.
3. s = [0, -2, 4]T.

Marking guide

- 4 points per class: 2 for the dot product, 1 for the correct bias, and 1 for the final score.
- Require exact values 0, -2, 4; accept row-vector notation when order is clear. Give method credit for one arithmetic error carried forward consistently.

b) [3 pts]

Answer: Class 2, Bird, because 4 is the largest score.

1. argmax([0, -2, 4]) = 2.
2. Map index 2 to Bird.

Marking guide

- 1 point for choosing the largest score; 2 for class 2/Bird. Accept consistent follow-through from part (a).

c) [5 pts]

Answer: Each row of W contains the weights or learned template for one class. It determines how each input feature contributes to that class's score. b contains one learned offset per class, shifting its score and the locations of decision boundaries without depending on the current input.

1. W has one row per class and one column per input feature.
2. Each b value is added to the corresponding class score.

Marking guide

- 3 points for W as learned feature weights/class templates, and 2 for b as a learned class offset.
- Accept algebraic, visual, or geometric explanations that are consistent with s = Wx + b.

d) [5 pts]

Answer: Its pairwise decision boundaries are hyperplanes: equality between two class scores gives a linear equation. Some data, such as XOR, require a nonlinear boundary. For XOR, (0,0) and (1,1) have one label while (0,1) and (1,0) have the other; no single straight line separates these two binary classes.

1. Set two scores equal to obtain $(w_i - w_j)\cdot x + (b_i - b_j) = 0$.
2. Explain a dataset whose classes cannot be separated by that type of boundary.
3. A nonlinear feature transform or a network with nonlinear activations can expand the model family.

Marking guide

- 2 points for linear/hyperplane limitations; 3 for a correct nonlinear example with explanation.
- Accept XOR, concentric rings, or an equivalent valid example. A drawing is optional; do not penalize its absence.

Question 4 - Understand the CV Pipeline [20 points]

a) [5 pts]

Answer: Under the default successful color read, img has NumPy shape (224, 224, 3): height 224, width 224, and 3 BGR channels. OpenCV's resize argument is (width, height); the square size makes the two spatial numbers equal.

1. cv2.imread uses color mode by default.
2. cv2.resize(img, (224, 224)) sets width and height to 224.
3. The operation does not remove color channels.

Marking guide

- 2 points for height 224, 2 for width 224, and 1 for 3 channels.
- Accept 224×224×3 with the order clearly stated. The question assumes that imread succeeds.

b) [5 pts]

Answer: The 3 BGR channels are combined into 1 grayscale intensity channel. The usual NumPy shape of gray is (224, 224), with no explicit singleton channel axis.

1. Conversion changes channel count, not spatial size.
2. A grayscale array commonly stores only height and width.

Marking guide

- 3 points for 3 → 1 channel; 2 for explaining intensity conversion or retaining spatial dimensions.
- Accept a conceptual 224×224×1 representation if it is distinguished from the actual 2D OpenCV array shape.

c) [5 pts]

Answer: Noise creates rapid intensity changes that can produce false edges. Gaussian smoothing reduces those high-frequency fluctuations before Canny computes gradients and selects edges. Too much blur can also suppress real fine edges.

1. Connect noise to unstable or false gradient responses.
2. Explain how smoothing makes edge detection less sensitive to noise.

Marking guide

- 2 points for identifying the noise/false-edge issue; 3 for the smoothing explanation.
- A correct practical noise-reduction explanation earns full marks; the blur trade-off is optional.

d) [5 pts]

Answer: No. Edges mainly describe boundaries and structure and do not retain the object's color. Preserve color images, or combine color-derived features with edge features, and compare alternatives on validation data.

1. Identify color as task-relevant information.
2. Show that grayscale edge extraction discards that information.
3. Retain color in the classifier input or feature set.

Marking guide

- 1 point for the decision, 3 for a correct task-based reason, and 1 for a viable color-preserving input.
- Accept RGB/BGR images, suitable color-space features, or combined color and edge features.

Question 5 - Design Your Own CV System [15 points]

a) [3 pts]

Answer: Assign one checked label to each image using the four classes. Store images with a label manifest, or class folders within each split. Remove duplicates and keep images of the same physical item together when creating independent train/validation/test sets.

1. Define a consistent four-class labeling scheme.
2. Use a reproducible image-to-label structure.
3. Prevent related images from crossing evaluation splits.

Marking guide

- 1 point for the four labels, 1 for a usable image/label organization, and 1 for a defensible organization or quality-control detail.
- Accept class folders or a manifest; there is no required directory format.

b) [3 pts]

Answer: One valid choice is 80% training and 20% testing: $0.8 \cdot 2000 = 1600$ training images and $0.2 \cdot 2000 = 400$ test images. Keep the test set untouched; use a validation subset or cross-validation within the training portion for model selection.

1. Choose a defensible split ratio.
2. Multiply each fraction by 2000.
3. Check that the counts add to 2000.

Marking guide

- 1 point for a valid split, 1 for correct training count, and 1 for correct testing count.
- Accept other justified ratios whose integer counts sum to 2000. A clearly explained three-way split is also defensible, although the prompt asks for train/test.

c) [3 pts]

Answer: Resize with aspect-ratio-preserving padding to a common input size, and normalize intensities using the model's expected convention. Mild denoising or a justified background crop are other possibilities.

1. Choose operations that standardize input or reduce irrelevant variation without erasing waste material cues.

Marking guide

- 1.5 points per relevant operation; accept any two justified task-appropriate choices.
- Augmentation may also be accepted if clearly used only on training images and label-preserving.

d) [3 pts]

Answer: Accuracy is a reasonable overall metric if the four classes are reasonably balanced, because it measures the fraction assigned the correct label. With imbalance, macro F1 or per-class precision and recall better expose poor minority-class performance; a confusion matrix shows which materials are confused.

1. Choose a metric appropriate to the stated or explicitly assumed class distribution.
2. Explain what it measures and why that matters.

Marking guide

- 1 point for a valid metric and 2 for a correct reason.
- Accept accuracy with a balanced-class assumption, macro F1, balanced accuracy, or relevant per-class metrics. No class proportions are given, so accuracy is not the only correct answer.

e) [3 pts]

Answer: Domain shift: the other phone may produce different lighting, color processing, sharpness, resolutions, viewpoints, or backgrounds from the training images. The classifier may have relied on those camera-specific patterns.

1. Identify a concrete difference in the input distribution.
2. Explain why the learned model may not generalize across it.

Marking guide

- 1 point for a plausible cause and 2 for its connection to worse performance.
- Accept camera/lighting/background shift, inconsistent preprocessing, overfitting, or leakage in the original dataset when explained.

Mock 2 — Sample-style: traffic signs and fruit sorting — Key

Question 1 - Image Processing & OpenCV [20 points]

a) [6 pts]

Answer: Resize with padding to a standard input size so batches align; apply a suitable intensity/illumination adjustment or model-specific normalization to reduce acquisition differences; use mild denoising to suppress sensor noise. Preserve sign color and small symbols.

1. Match resolution, illumination, and noise to suitable operations.
2. Explain the information each operation should preserve.

Marking guide

- 2 points per appropriate operation and reason.
- Accept other justified choices such as a reliable sign crop; require three distinct operations.

b) [4 pts]

Answer: One channel reduces memory and computation. However, grayscale can remove color distinctions such as a red border versus a blue background that are useful for recognizing sign categories.

1. Compare channel count.
2. Connect the lost information to the sign classification task.

Marking guide

- 2 points for a valid advantage; 2 for a task-specific disadvantage. Accept equivalent explanations.

c) [5 pts]

Answer: Default imread returns BGR, so replace COLOR_RGB2GRAY with COLOR_BGR2GRAY. GaussianBlur needs positive odd specified kernel sizes, so replace (6, 6) with (5, 5).

```
img = cv2.imread("sign.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
blur = cv2.GaussianBlur(gray, (5, 5), 0)
```

1. Correct the color-order mismatch.
2. Choose a valid odd Gaussian kernel.
3. Optionally check img is not None before conversion.

Marking guide

- 2 points for the color correction and reason; 2 for the kernel correction and reason; 1 for usable code.
- Accept another valid positive odd kernel.

d) [5 pts]

Answer: No. Strong smoothing can erase a thin digit stroke, an arrow, or a small boundary detail, making distinct signs look alike. Use only as much denoising as validation results justify.

1. Name a discriminative sign feature.
2. Explain how smoothing can remove it and cause an error.

Marking guide

- 1 point for rejecting the universal claim; 3 for a specific feature and mechanism; 1 for a justified evaluation approach.
- Accept any concrete sign feature and correct explanation.

Question 2 - Image Classification [20 points]

a) [4 pts]

Answer: A: Stop; B: Yield; C: Yield; D: Speed limit.

1. A: 1.2 is highest.
2. B: 2.1 is highest.
3. C: 1.9 is highest.
4. D: 1.0 is highest.

Marking guide

- 1 point per correct class.

b) [4 pts]

Answer: A, B, and D are correct. Accuracy = 3/4 = 75%.

1. Count 3 correct labels.
2. Divide by 4 evaluated images.

Marking guide

- 1 point for the formula, 2 for the correct numerator/denominator, and 1 for 75% or 0.75.

c) [4 pts]

Answer: C is the only error: predicted Yield, true Speed limit.

1. Compare the largest score in C with the label in the last column.

Marking guide

- 2 points for C; 1 each for predicted Yield and true Speed limit.

d) [4 pts]

Answer: No. A raw class score is not constrained to [0,1] or normalized across classes. Scores such as 2.1 can rank classes; an explicit transformation such as softmax produces a probability distribution.

1. Identify 2.1 as a raw score.
2. State probability bounds and normalization.

Marking guide

- 1 point for no; 3 for the correct score/probability explanation. Softmax need not be calculated.

e) [4 pts]

Answer: Overfitting is likely. Use a remedy such as valid training augmentation, weight decay, or validation-based early stopping. Check distribution differences if the test images were collected differently.

1. Relate the large gap to generalization.
2. Tie a specific remedy to the proposed cause.

Marking guide

- 2 points for a justified cause; 2 for a relevant remedy. Accept a supported domain-shift diagnosis with a matched remedy.

Question 3 - Linear Classifier [25 points]

a) [12 pts]

Answer: s = [2, -3, 3.5]T. Stop: 1·3 + (-1)·2 + 1 = 2. Yield: (-2)·3 + 2·2 - 1 = -3. Speed limit: 0.5·3 + 1·2 + 0 = 3.5.

1. Wx = [1, -2, 3.5]T.
2. Add b to get [2, -3, 3.5]T.

Marking guide

- 4 points per class: 2 for the dot product, 1 for bias, 1 for the score.
- Require 2, -3, and 3.5 exactly; accept a consistent arithmetic carry-through for method marks.

b) [3 pts]

Answer: Class 2, Speed limit, because 3.5 is highest.

1. Select argmax and map the index to its name.

Marking guide

- 1 point for selecting the maximum; 2 for class 2/Speed limit. Accept follow-through from part (a).

c) [5 pts]

Answer: W is 3×2: three classes, two input features. b is a length-3 vector. Each row of W gives the feature weights/template used to score one class; the corresponding bias is its input-independent offset.

1. Use one output per class and one weight per feature per class.
2. Explain the row-to-class correspondence.

Marking guide

- 2 points for W shape, 1 for b shape, and 2 for a correct semantic explanation.

d) [5 pts]

Answer: No. Opposite corners share a label, so no straight line places both of one label on one side and both of the other label on the other. Nonlinear features or a network with a nonlinear hidden layer can represent such a boundary.

1. List XOR labels: (0,0),(1,1) in one class; (0,1),(1,0) in the other.
2. Explain the straight-line limitation.

Marking guide

- 1 point for no; 3 for the valid geometric reason; 1 for identifying a nonlinear extension. A drawing is optional.

Question 4 - Understand the CV Pipeline [20 points]

a) [5 pts]

Answer: (96, 160, 3). The size argument sets width to 160 and height to 96; the default color read has 3 channels.

1. Read resize arguments as (width, height).
2. Write array shape as (height, width, channels).

Marking guide

- 2 points for height 96, 2 for width 160, and 1 for 3 channels.

b) [5 pts]

Answer: Each has one channel and a 2D array shape (96,160). Grayscale conversion and Canny retain these spatial dimensions.

1. Grayscale removes the color channel axis.
2. Canny produces a single-channel edge map.

Marking guide

- 2 points for gray, 2 for edges, and 1 for retained spatial dimensions.

c) [5 pts]

Answer: It suppresses noise-induced intensity fluctuations, reducing spurious gradients and false edges. Mild smoothing can give more stable sign boundaries; excessive blur can suppress real details.

1. Connect sensor noise with unstable gradients.
2. Explain the effect of smoothing.

Marking guide

- 2 points for noise/false edges and 3 for the relevant smoothing explanation.

d) [5 pts]

Answer: It may be insufficient because the silhouettes can produce similar edges while the class distinction lies in color. Keep color images or combine color-space features with edges.

1. Identify the information removed by the pipeline.
2. Propose an input that preserves it.

Marking guide

- 1 point for the decision; 3 for the color-information reason; 1 for an appropriate improvement. Accept valid color-preserving alternatives.

Question 5 - Design a Fruit-Sorting System [15 points]

a) [3 pts]

Answer: Use consistent four-class labels in folders or a manifest; remove duplicates; keep all photographs of a physical fruit in one split so the model cannot see that fruit in training and testing.

1. Record class and physical-item identity.
2. Assign related photos as a group.

Marking guide

- 1 point for image/label organization, 1 for the four-class scheme, and 1 for keeping related photos together. Accept equivalent leakage prevention.

b) [3 pts]

Answer: Training: 0.8·3200 = 2560. Testing: 0.2·3200 = 640. Total = 3200.

1. Multiply each given fraction by 3200.
2. Check the sum.

Marking guide

- 1 point for the calculation method; 1 each for 2560 and 640. Exact integer counts are required.

c) [3 pts]

Answer: Resize with padding to standardize shape, and apply the required intensity normalization. Preserve color because ripeness or fruit type may depend on it. A reliable background crop is another option.

1. Choose two distinct operations.
2. Retain class-relevant appearance.

Marking guide

- 1.5 points for each valid operation. Accept other justified choices; no specific pair is required.

d) [3 pts]

Answer: Macro F1 gives each class equal weight and combines precision with recall, making weak Other performance more visible than overall accuracy. Report a confusion matrix or per-class metrics to locate errors.

1. Identify how a majority can dominate overall accuracy.
2. Choose a metric that exposes minority-class performance.

Marking guide

- 1 point for a suitable metric; 2 for an imbalance-related reason. Accept macro F1, balanced accuracy, or relevant per-class precision/recall.

e) [3 pts]

Answer: Domain shift in backgrounds, lighting, occlusion, or camera settings can change the appearance distribution. Studio-only training may encourage reliance on background cues.

1. Name a concrete environmental change.
2. Connect it to a learned shortcut or unseen variation.

Marking guide

- 1 point for a plausible change; 2 for its correct connection to poor generalization. Accept any well-explained cause.

Mock 3 — Sample-style: industrial inspection — Key

Question 1 - Image Processing & OpenCV [20 points]

a) [6 pts]

Answer: Resize or pad to a consistent input size; apply a justified illumination/intensity adjustment so lighting is less variable; use a median filter to remove isolated salt-and-pepper outliers while preserving edges reasonably well. Avoid destroying thin cracks or rust color.

1. Map each operation to a stated acquisition problem.
2. Connect median filtering to isolated extreme pixel values.

Marking guide

- 2 points per valid operation with a suitable reason.
- Accept other justified operations. A Gaussian filter is less targeted to salt-and-pepper noise but can earn credit if its trade-off is explained.

b) [4 pts]

Answer: Grayscale can reduce computation and retain crack shape/contrast. It can also remove orange-brown color evidence that helps distinguish rust from normal metal.

1. Identify an efficiency or structural benefit.
2. Identify a color cue that would be lost.

Marking guide

- 2 points for a valid benefit and 2 for a relevant limitation. Accept equivalent task-based answers.

c) [5 pts]

Answer: IMREAD_GRAYSCALE already returns a single-channel image, so BGR2GRAY is inappropriate and unnecessary. medianBlur also requires an odd kernel size greater than 1; change 4 to 5.
gray = cv2.imread("bolt.png", cv2.IMREAD_GRAYSCALE)
blur = cv2.medianBlur(gray, 5)

1. Use the already grayscale image directly.
2. Replace the even median kernel size with a valid odd size.
3. Alternatively read in color first, then use BGR2GRAY and medianBlur with an odd kernel.

Marking guide

- 2 points for the channel/conversion correction; 2 for the kernel correction; 1 for coherent code.
- Accept a color read followed by a correct conversion; accept a valid odd median kernel appropriate to the image type.

d) [5 pts]

Answer: A real crack can be a small dark region. An operation that removes every such region may erase positive-class evidence along with noise, making defective bolts appear normal. Distinguish noise from useful structure and verify recall on validation images.

1. Identify overlap between the target and the proposed removal rule.
2. Explain the resulting false negatives.

Marking guide

- 1 point for the decision; 3 for the correct crack/false-negative explanation; 1 for a task-relevant validation suggestion. Accept equivalent reasoning.

Question 2 - Image Classification [20 points]

a) [4 pts]

Answer: A: Normal; B: Rust; C: Normal; D: Rust.

1. Largest scores are respectively 1.4, 1.9, 1.8, and 1.1.

Marking guide

- 1 point per correct predicted label.

b) [4 pts]

Answer: Only A and B are correct. Accuracy = 2/4 = 50%.

1. Compare predictions with true labels.
2. Divide 2 correct predictions by 4 images.

Marking guide

- 1 point for the accuracy formula, 2 for the correct numerator/denominator, and 1 for 50% or 0.5.

c) [4 pts]

Answer: C: true Crack, predicted Normal. D: true Crack, predicted Rust.

1. Check C and D against the last column.

Marking guide

- 2 points per error correctly identified with both labels.

d) [4 pts]

Answer: A score is an unconstrained numerical preference from the model; only relative score order is needed for argmax. Probabilities represent a normalized distribution and must lie in [0,1].

1. Distinguish model score scale from probability constraints.

Marking guide

- 2 points for correctly describing raw scores; 2 for the probability constraint.

e) [4 pts]

Answer: Overfitting is plausible; use training augmentation, regularization, or validation-based early stopping. If lighting/camera conditions differ across splits, collect representative training images or standardize the preprocessing to address domain shift.

1. State one cause.
2. Choose a remedy that targets that cause.

Marking guide

- 2 points for a justified diagnosis; 2 for a relevant improvement. The gap alone does not establish the cause.

Question 3 - Linear Classifier [25 points]

a) [12 pts]

Answer: Normal: 2·(-2) + 1·3 + 0 = -1. Crack: (-1)·(-2) + 2·3 - 2 = 6. Rust: 1·(-2) + (-2)·3 + 1 = -7. Thus s = [-1, 6, -7]T.

1. Wx = [-1, 8, -8]T.
2. Add [0, -2, 1]T to get [-1, 6, -7]T.

Marking guide

- 4 points per score: 2 for the dot product, 1 for adding bias, 1 for the final value.
- Require exact integer scores; allow consistent follow-through for arithmetic errors.

b) [3 pts]

Answer: Class 1, Crack; choose the highest of the three scores, 6.

1. argmax([-1,6,-7]) = 1.

Marking guide

- 1 point for the rule; 2 for class 1/Crack. Accept consistent follow-through.

c) [5 pts]

Answer: Changing a W row changes how input features influence one class score, so its effect depends on x. Changing the corresponding bias adds the same offset to that class score for every input, shifting its preference and pairwise boundaries.

1. Identify input-dependent feature effects from W.
2. Identify the input-independent offset from b.

Marking guide

- 3 points for the weight-row interpretation and 2 for the bias interpretation. Accept correct algebraic/geometric explanations.

d) [5 pts]

Answer: A ring surrounds the inner class, so a straight-line binary boundary cannot separate the full ring from its center. A nonlinear feature such as distance from the center, or a nonlinear model, can represent this structure.

1. Connect linear scores with straight boundaries in two dimensions.
2. Explain the surrounding-ring geometry.

Marking guide

- 2 points for the linear-boundary limitation; 2 for a correct ring explanation; 1 for a suitable nonlinear extension. Accept equivalent reasoning.

Question 4 - Understand the CV Pipeline [20 points]

a) [5 pts]

Answer: (64, 128, 3). OpenCV reads the resize size as (width, height).

1. Width is 128, height is 64, color channels remain 3.

Marking guide

- 2 points for height 64, 2 for width 128, and 1 for channels 3.

b) [5 pts]

Answer: Both arrays have shape (64,128) and one channel. gray may have uint8 intensities from 0 to 255. binary contains only 0 and 255.

1. Grayscale conversion retains spatial size.
2. Binary threshold maps intensities to two output values.

Marking guide

- 2 points for the shapes/channel count; 1 for gray’s range; 2 for binary values 0 and 255.

c) [5 pts]

Answer: An illumination change shifts measured intensities, so the same material may fall on opposite sides of 120. Illumination normalization or a suitable adaptive threshold can help; validate the result because material and defect cues must remain distinguishable.

1. Connect lighting with changed pixel intensities.
2. Explain incorrect foreground/background decisions.
3. Propose a relevant adjustment.

Marking guide

- 3 points for the correct failure mechanism; 2 for a justified remedy. Accept controlled lighting, normalization, or suitable threshold selection.

d) [5 pts]

Answer: It can be insufficient because thresholding destroys color and most intensity detail. Keep the color image or color-space features, possibly alongside structural features.

1. Identify the relevant color distinction.
2. Explain which information binary thresholding discards.

Marking guide

- 1 point for the decision, 3 for a correct explanation, and 1 for a color-preserving alternative.

Question 5 - Design an Inspection System [15 points]

a) [3 pts]

Answer: Maintain four consistent class labels in folders or a manifest and include part IDs. Keep all images of each part within one split, rather than putting near-identical part photos in both training and evaluation.

1. Organize image-to-label mapping.
2. Split by physical-part identity.

Marking guide

- 1 point for a usable organization, 1 for class consistency, and 1 for avoiding related-image leakage.

b) [3 pts]

Answer: Training: 1050; validation: 225; test: 225. Counts sum to 1500.

1. 0.70·1500 = 1050.
2. 0.15·1500 = 225 for each evaluation subset.

Marking guide

- 1 point per exact count.

c) [3 pts]

Answer: Resize with padding, and mild median denoising when impulse noise is present. Model-specific intensity normalization or a reliable part crop are also reasonable. Preserve fine cracks and rust color.

1. Choose operations relevant to the task's image variation.

Marking guide

- 1.5 points for each valid choice. Accept any two distinct task-appropriate operations.

d) [3 pts]

Answer: Crack recall = TP/(TP+FN) matters because it measures the fraction of genuinely cracked parts detected. It directly exposes missed cracks; precision and inspection costs should also be monitored.

1. Identify false negatives as missed cracks.
2. Relate recall to detecting the true positive class.

Marking guide

- 1 point for crack recall, 1 for the correct interpretation/formula, and 1 for the cost-based explanation.
- Accept an appropriate cost-sensitive metric if the rationale is explicit.

e) [3 pts]

Answer: Background domain shift: the model may have associated the original conveyor background with a label and now sees different texture/color. Lighting, camera angle, or motion blur could also differ.

1. Identify one concrete deployment difference.
2. Explain how the classifier's reliance on it can hurt generalization.

Marking guide

- 1 point for the cause; 2 for its relevant explanation. Accept any well-supported acquisition or distribution shift.

Mock 4 — Expanded lecture coverage: losses, learning, CNNs — Key

Question 1 - kNN and Evaluation [20 points]

a) [6 pts]

Answer: A: 3; B: 1; C: 2; D: 4; E: 1.

1. A: $|1-0|+|2-0|=3$.
2. B: $|1-2|+|2-2|=1$.
3. C: $|1-1|+|2-4|=2$.
4. D: $|1-4|+|2-1|=4$.
5. E: $|1-0|+|2-2|=1$.

Marking guide

- 1 point for each exact distance and 1 for showing a correct distance calculation or formula.

b) [4 pts]

Answer: k=1: B (tie with E resolved by ID), so Paper. k=3: B, E, C; all are Paper, so Paper.

1. Sort by distance, then ID: B, E, C, A, D.
2. Use the nearest label for k=1 and majority vote for k=3.

Marking guide

- 2 points for each correct neighbor selection and label. Follow the stated tie convention.

c) [5 pts]

Answer: Choose hyperparameters using a validation set or cross-validation on the training portion. Reserve the independent final test for evaluation after choices are fixed; selecting the best test result makes the reported performance optimistically biased.

1. Separate fitting, model selection, and final evaluation.
2. Explain how repeated test-based choices leak information into model selection.

Marking guide

- 2 points for validation/cross-validation, 2 for test-set selection bias, and 1 for a correct final-evaluation protocol. Accept equivalent explanations.

d) [5 pts]

Answer: Raw pixel distances can be dominated by lighting, shifts, or backgrounds, and high-dimensional space requires many examples for useful coverage. A learned feature representation can make distances more semantically meaningful; reducing dimensionality or controlling acquisition may also help.

1. Identify one concrete raw-pixel/high-dimensional difficulty.
2. Connect an improvement to that difficulty.

Marking guide

- 3 points for a correct difficulty with explanation; 2 for a matched improvement.
- Accept curse-of-dimensionality, shift sensitivity, nuisance variation, or brute-force distance cost with an appropriate remedy.

Question 2 - Scores and Losses [20 points]

a) [4 pts]

Answer: Class 0 is predicted because its score 2 is largest. It is incorrect because the true class is 1.

1. Take argmax over [2,1,0].
2. Compare index 0 with index 1.

Marking guide

- 2 points for predicted index 0 and 2 for the correctness comparison.

b) [6 pts]

Answer: p ≈ [0.665,0.245,0.090]. LCE = -ln(0.244728471) ≈ 1.408.

1. For stability, subtract 2: exp scores [1,e^-1,e^-2].
2. Divide by 1+e^-1+e^-2 ≈ 1.503214724.
3. Use the true-class probability p1 ≈ 0.244728471, not the largest probability.

Marking guide

- 1 point per probability (3 total), 1 for normalization, 1 for using p1, and 1 for LCE≈1.408.
- Accept probabilities within 0.001 and loss within 0.002; accept exact exponential/log expressions. Avoid computing loss from over-rounded probabilities when exact values are available.

c) [6 pts]

Answer: For j=0: max(0,2-1+1)=2. For j=2: max(0,0-1+1)=0. LH=2+0=2.

1. Exclude the true-class term.
2. Use margin 1 for each other class.
3. Sum both nonnegative terms.

Marking guide

- 2 points per correct term, 1 for excluding j=y, and 1 for exact total 2.

d) [4 pts]

Answer: Highest score alone does not imply zero hinge loss: the correct score must exceed each other score by at least the margin 1. Softmax cross-entropy is greater than zero for finite scores with more than one class, because the correct probability is strictly less than 1; it approaches zero as that probability approaches 1.

1. Distinguish argmax correctness from satisfying a margin.
2. Distinguish high probability from probability exactly 1.

Marking guide

- 2 points for the hinge margin condition; 2 for the finite-score cross-entropy condition. Accept equivalent mathematically correct explanations.

Question 3 - Gradient Descent and Branching [25 points]

a) [5 pts]

Answer: yhat=0.25·2+0=0.5. L=0.5(0.5-1)^2=0.125.

1. Compute the linear prediction.
2. Compute the residual -0.5, square it, and multiply by 0.5.

Marking guide

- 2 points for exact prediction 0.5; 3 for exact loss 0.125 with the stated one-half factor.

b) [8 pts]

Answer: ∂L/∂yhat=yhat-t=-0.5. ∂yhat/∂w=x=2 and ∂yhat/∂b=1. Therefore ∂L/∂w=-1 and ∂L/∂b=-0.5.

1. Differentiate 0.5(yhat-t)^2.
2. Multiply the upstream residual by each local derivative.

Marking guide

- 2 points for the residual derivative; 2 for local derivatives; 2 each for the exact parameter gradients.

c) [6 pts]

Answer: wnew=0.25-0.1(-1)=0.35. bnew=0-0.1(-0.5)=0.05. yhatnew=0.35·2+0.05=0.75. Lnew=0.5(0.75-1)^2=0.03125.

1. Subtract η times each old gradient.
2. Evaluate using both new parameters.
3. The loss decreases from 0.125 to 0.03125 for this step.

Marking guide

- 2 points for the parameter updates, 2 for prediction 0.75, and 2 for loss 0.03125.
- Accept decimal results within 1e-6 or exact fractions. Do not recompute the second gradient after updating the first parameter.

d) [6 pts]

Answer: The u·u branch contributes u+u=4 and the 3·u branch contributes 3. Since the same u influences both terms, df/du=4+3=7.

1. Treat the two uses of u in multiplication as separate paths; each contributes u=2.
2. Add the third contribution, 3.
3. Sum all paths that reach the shared input.

Marking guide

- 2 points for derivative 4 of u·u, 1 for derivative 3 of 3·u, 2 for summing branches, and 1 for exact total 7.

Question 4 - CNN Shapes and Parameters [20 points]

a) [6 pts]

Answer: After layer 1/ReLU: 32×32×8. After pooling: 16×16×8.

1. Conv: (32-3+2·1)/1+1=32; output channels=8.
2. ReLU changes values but not shape.
3. Pool: (32-2)/2+1=16; channels stay 8.

Marking guide

- 3 points for each correct shape with supporting reasoning. Require exact integer dimensions.

b) [5 pts]

Answer: 14×14×16, because (16-3)/1+1=14 and there are sixteen filters.

1. Use pooled spatial size 16 as the input.
2. Output depth equals filter count.

Marking guide

- 3 points for spatial calculation and 2 for correct channels 16.

c) [5 pts]

Answer: Layer 1: 8·(3·3·3+1)=224. Layer 2: 16·(3·3·8+1)=1168. ReLU and max pooling contribute no trainable parameters.

1. Each filter spans all input channels.
2. Include one bias per output filter.
3. Count weights once, because they are shared across positions.

Marking guide

- 2 points for 224, 2 for 1168, and 1 for a correct parameter-count method including biases. Exact counts are required.

d) [4 pts]

Answer: Local connectivity uses spatially neighboring pixels to detect patterns. Weight sharing reuses one filter at every position, reducing parameter count and making a shifted feature produce a shifted response, subject to borders/stride. A fully connected layer uses separate weights for every input-to-unit connection.

1. Link locality to spatial structure.
2. Link sharing to reuse, parameter economy, and shifted responses.

Marking guide

- 2 points for locality and 2 for weight sharing. Accept equivalent explanations; do not require a claim of perfect translation invariance.

Question 5 - Build and Evaluate a Leaf Classifier [15 points]

a) [5 pts]

Answer: Nominal counts: 5250 training, 1125 validation, 1125 test. Keep all views of a plant in one split; grouped splitting may slightly change exact image counts to maintain independence. Use validation for model selection and test for final evaluation.

1. Multiply fractions by 7500.
2. Treat plant identity as a grouping unit.
3. Assign distinct roles to validation and test.

Marking guide

- 1 point for each nominal count, 1 for plant-group separation, and 1 for correct validation/test roles.

b) [5 pts]

Answer: Overfitting or distribution mismatch is plausible. Use valid augmentation that represents realistic camera/light variation, and stronger regularization or validation-based early stopping. Also inspect labels and split independence rather than assuming that the gap has only one cause.

1. Connect the accuracy gap with weak generalization.
2. Choose two interventions and state how each could help.

Marking guide

- 1 point for a plausible cause; 2 points per action with a correct reason.
- Accept improved representative data, appropriate model-capacity changes, label auditing, or other matched remedies.

c) [5 pts]

Answer: MobileNet is a reasonable candidate because depthwise separable convolutions reduce computation; a small EfficientNet is another option. Compare validation accuracy/per-class metrics and measured latency, memory, and energy on the target phone, then evaluate the selected system on the untouched test.

1. Identify an efficiency-oriented architecture.
2. Measure the trade-off on the deployment hardware.
3. Avoid choosing the architecture by repeatedly using the final test.

Marking guide

- 2 points for a suitable family and relevant reason; 2 for accuracy/efficiency comparison; 1 for an independent final test.
- Accept another justified small architecture; no architecture is automatically best for every device.

Mock 5 — Expanded lecture coverage: filtering, backprop, architecture — Key

Question 1 - Image Processing Decisions [20 points]

a) [5 pts]

Answer: (1+2+3+4+5+6+7+8+9)/9 = 45/9 = 5.

1. Multiply each patch value by 1/9.
2. Sum the nine products.

Marking guide

- 2 points for the averaging method, 2 for sum 45, and 1 for exact output 5.

b) [5 pts]

Answer: Sorted values: [9,9,10,10,10,10,11,11,255]. The fifth value is 10. The median ignores the magnitude of the isolated 255 outlier rather than allowing it to dominate the local estimate.

1. Sort the nine values.
2. Choose the fifth value, not their average.
3. Relate robustness to impulse noise.

Marking guide

- 2 points for sorting/selecting the median, 1 for exact output 10, and 2 for the outlier explanation.

c) [5 pts]

Answer: The unbounded results are [20,116,212,402]; clipping gives [20,116,212,255]. Compute in a sufficiently wide or floating type and clip before casting so out-of-range values do not wrap or overflow.

1. Multiply and add in a wide numeric type.
2. Clamp the final value to the valid range.
3. Cast to uint8 after clamping.

Marking guide

- 3 points for the four correct outputs, 1 for clipping 402 to 255, and 1 for the overflow/wrap explanation. Exact values are required.

d) [5 pts]

Answer: Downsampling lowers the sampling rate. High-frequency patterns can then appear as false lower-frequency structures, such as moiré. Apply an appropriate low-pass/anti-alias filter before decimation so frequencies above the new sampling limit are suppressed.

1. Connect lower sample density to unresolved high frequencies.
2. Explain the purpose of prefiltering.

Marking guide

- 3 points for the correct aliasing mechanism; 2 for an appropriate low-pass step. Accept a correctly described anti-aliased resize/interpolation method.

Question 2 - A Two-Layer Network [20 points]

a) [6 pts]

Answer: $z1=1\cdot1+1\cdot(-2)+0.5=-0.5$; $z2=(-1)\cdot1+(-2)\cdot(-2)-0.5=2.5$. Therefore $z=[-0.5,2.5]T$ and $h=[0,2.5]T$.

1. Compute each affine hidden value.
2. ReLU replaces the negative entry with zero and keeps the positive entry.

Marking guide

- 2 points for each z entry and 2 for the exact ReLU output.

b) [4 pts]

Answer: $s0=2\cdot0+(-1)\cdot2.5+0.5=-2$. $s1=1\cdot0+2\cdot2.5-0.5=4.5$. Class 1 is predicted.

1. $s=[-2,4.5]T$.
2. The maximum is at index 1.

Marking guide

- 1.5 points per exact score and 1 for class 1. Accept consistent follow-through from (a).

c) [6 pts]

Answer: s=W2(W1x+b1)+b2=(W2W1)x+(W2b1+b2). Wprime=[[3,4],[-1,-3]] and bprime=[2,-1]T. Stacking these affine layers without a nonlinear activation still gives a single affine classifier, so depth alone does not create nonlinear boundaries.

1. Multiply W2W1: first row [3,4], second row [-1,-3].
2. Compute W2b1=[1.5,-0.5]T, then add b2.
3. Identify the unchanged affine model family.

Marking guide

- 2 points for the derivation, 2 for exact Wprime/bprime, and 2 for the implication. Accept equivalent algebraic explanations.

d) [4 pts]

Answer: Its ReLU local derivative is 0, so this example sends zero gradient through that branch. It is not necessarily permanently dead: other inputs may make its preactivation positive. A dead unit stays inactive across the relevant inputs and receives no useful activation gradient.

1. Apply the derivative at -0.5.
2. Distinguish one inactive example from persistent inactivity.

Marking guide

- 2 points for zero gradient and 2 for the correct distinction. No convention at exactly zero is needed here.

Question 3 - Matrix Backpropagation [25 points]

a) [5 pts]

Answer: s=[2.5,-4.5]T. W has shape 2×2; x and g are 2×1 column vectors.

1. s0=1·2+0·(-1)+0.5=2.5.
2. s1=(-1)·2+2·(-1)-0.5=-4.5.

Marking guide

- 2 points for scores, 1 for W shape, and 1 each for x and g shapes. Equivalent length-2 array notation is accepted if column-vector operations are clear.

b) [8 pts]

Answer: $\partial L/\partial W = gx^T = [[3],[-2]]\cdot[[2,-1]] = [[6,-3],[-4,2]]$. $\partial L/\partial b = g = [3,-2]^T$.

1. Gradient entry (i,j) is $g_i \cdot x_j$.
2. Adding b to s gives identity local derivatives.
3. Gradient shapes match W and b.

Marking guide

- 2 points for gx^T and its shape, 1 point per correct matrix entry (4 total), and 2 for db. Exact values are required.

c) [6 pts]

Answer: $\partial L/\partial x = W^T g = [[1,-1],[0,2]]\cdot[3,-2]^T = [5,-4]^T$. Each input contributes to multiple output scores, so its gradient sums the corresponding W column weighted by g.

1. $dx1 = 1\cdot3 + (-1)\cdot(-2) = 5$.
2. $dx2 = 0\cdot3 + 2\cdot(-2) = -4$.
3. W^T maps the output-gradient column back to the input-gradient column.

Marking guide

- 2 points for $W^T g$, 1.5 points per exact gradient entry, and 1 for a correct shape/contribution explanation.

d) [6 pts]

Answer: $[16.500060 - 16.499940]/(2\cdot0.00001) = 0.000120/0.000020 = 6$. This agrees with $\partial L/\partial W[0,0] = 6$.

1. Use the two-sided central-difference formula.
2. Calculate numerator and denominator separately.
3. Compare with the matching weight entry, not dx or db.

Marking guide

- 2 points for the formula, 2 for numerical result 6, and 2 for the correct analytic comparison.
- Accept within 1e-6 given the stated rounded loss values. This validates this coordinate for this example; it does not prove every gradient implementation correct.

Question 4 - CNN Architecture [20 points]

a) [6 pts]

Answer: Conv A: 24×24×6. Pool: 12×12×6. Conv B: 12×12×12. Global average pooling: 1×1×12, or a vector of 12 features.

1. Conv A: 28-5+1=24.
2. Pool: (24-2)/2+1=12.
3. Conv B: 12-3+2·1+1=12.
4. Average each feature map over its spatial positions.

Marking guide

- 1.5 points per correct stage. Accept the GAP vector representation when its 12 features are stated.

b) [6 pts]

Answer: Conv A: 6·(5·5·1+1)=156. Conv B: 12·(3·3·6+1)=660. FC: 12·3+3=39. Total=156+660+39=855. Pooling and global average pooling add no learned parameters.

1. Include full input-channel depth in each filter.
2. Include one bias per output.
3. GAP gives the FC head exactly 12 inputs.

Marking guide

- 1.5 points per exact layer count and 1.5 for exact total 855. Give method credit for correct follow-through.

c) [4 pts]

Answer: Both have a 5×5 receptive field. Two 3×3 layers use $2 \cdot 9C^2 = 18C^2$ weights versus $25C^2$ for one 5×5 layer, and can include an additional intermediate nonlinearity.

1. Two 3-wide layers see 1+2·(3-1)=5 input positions per axis.
2. Count both sets of 3×3 weights under the stated equal-channel assumption.

Marking guide

- 2 points for receptive field 5×5; 2 for correct comparison $18C^2$ versus $25C^2$. The extra nonlinearity explanation is useful but optional.

d) [4 pts]

Answer: A residual block adds $y=F(x)+x$ before any final activation. Elementwise addition requires matching spatial and channel shapes. If the main path changes resolution or channels, a suitable projection, often a stride/channel-adjusting 1×1 convolution, can match the skip shape.

1. State the identity addition.
2. Check shape compatibility before addition.

Marking guide

- 2 points for the relation, 1 for matching-shape requirement, and 1 for a suitable projection. Accept equivalent valid shape-matching methods.

Question 5 - Evaluation Under Imbalance and Shift [15 points]

a) [5 pts]

Answer: $\text{Precision}=TP/(TP+FP)=12/40=0.30=30\%$. $\text{Recall}=TP/(TP+FN)=12/20=0.60=60\%$.

1. Use predicted-positive count 40 for precision.
2. Use actual-positive count 20 for recall.

Marking guide

- 2.5 points per metric: formula, denominator, and correct value. Accept exact fractions; decimals within 0.001.

b) [5 pts]

Answer: $\text{Model accuracy}=(TP+TN)/1000=964/1000=96.4\%$. Always Normal gives $980/1000=98\%$. The baseline detects no faulty items despite higher accuracy; minority-class recall and precision, or an explicit cost measure, reveal this weakness.

1. Count all correct predictions for accuracy.
2. Compute the majority-class baseline.
3. Compare the positive-class behavior, not only overall correctness.

Marking guide

- 2 points for 96.4%, 1 for 98%, and 2 for the correct imbalance explanation. Accept equivalent cost/per-class reasoning.

c) [5 pts]

Answer: A camera or lighting domain shift, or inconsistent input preprocessing, may change the image distribution. Check the exact color order/normalization and collect representative images for a separate development/validation process. Do not keep tuning on the final test; if it informs changes, use a fresh independent test to evaluate the revised system.

1. Name a specific distribution or preprocessing difference.
2. Match a data/processing remedy to it.
3. Keep final evaluation independent of subsequent selection.

Marking guide

- 1.5 points for a plausible cause, 1.5 for a matched remedy/check, and 2 for the correct test-set protocol.
- Accept domain shift, preprocessing mismatch, or a supported alternate explanation; multiple remedies are possible.